

THIRUKUMARAN T

Madurai, Tamil Nadu | +91-9944735972 | [Email](#) |

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY:

M.Pharm in Pharmaceutical Biotechnology (CGPA 9.42/10, Tata Trusts Scholar) with hands-on expertise in single-cell transcriptomics, NGS data analysis, and bioinformatics pipeline development. Experienced in building reproducible scRNA-seq workflows using R, Seurat, and Docker, with applied knowledge of bioanalytical assay development from industry training at a WHO-prequalified vaccine manufacturer. Passionate about bridging computational biology and translational research to accelerate drug discovery and precision medicine.

PROJECTS:

Tumour microenvironment characterisation in glioblastoma (scRNA-seq analysis):

- Characterised the full cellular landscape of human glioblastoma using single-cell transcriptomics: resolved 12 cell populations (2,847 cells; 24,381 genes; 10x Genomics) including clinically relevant malignant subtypes -OPC-like, mesenchymal-like, and hypoxia-associated - each with distinct therapeutic implications.
- Developed a reproducible pipeline (R, Seurat, Docker, GitHub Actions) translatable to clinical NGS datasets for tumour subtype stratification.
- Published detailed **research blog**: [<https://fromreadstoreultsttk.blogspot.com/>] | **GitHub** - [<https://github.com/TTK-Pharma/NGS-data-analysis>]

Tools: R (Seurat, DESeq2, Bioconductor), GitHub

Biopharmaceutical Characterization and Functional Potency Assessment of Poliovirus-Like Particles (PVLP)

- Conducted independent academic characterisation of 5 PVLP vaccine batch samples, evaluating host cell protein (HCP), host cell DNA (HCD), and protein integrity markers as part of M. Pharm dissertation work.
- Analyzed and interpreted experimental data across multiple assays; identified quality parameters within WHO and regulatory compliance guidelines.
- Published more informative **research blog**: [<https://thirukumaranrtm.blogspot.com/>]

Tools & Techniques: ELISA, PicoGreen Assay, BCA Assay, SDS-PAGE, SNT, Cell Culture Techniques

CLI tool for reproducible scRNA-seq analysis | **Tools:** R, Seurat, Docker | [In Development]

- Developed a configurable CLI pipeline for scRNA-seq analysis (R, Seurat, Docker) designed for reproducibility across 10x Genomics datasets — applicable to clinical tumour profiling workflows.
- Implements QC, normalisation, clustering, and visualisation via user-defined parameters.

Molecular characterization of APOE gene SNPs in Alzheimer's Disease

- Analyzed APOE SNPs using dbSNP, 1000 Genomes, and ADSP datasets to study Alzheimer's disease risk.
- Performed functional prediction using PolyPhen-2, SIFT, and FATHMM to identify deleterious variants.
- Identified high-risk variants including $\epsilon 4$ (rs429358) associated with increased AD susceptibility.

SKILLS:

Bioinformatics & Genomics:

- scRNA-seq analysis
- Differential Expression Analysis
- Variant Analysis
- Sequence Analysis
- Molecular Docking

Programming & Data Analysis:

- R
- Python
- SQL
- Docker, Git, Linux

Bioanalytical assay:

- HCP (Host Cell Protein) assay
- HCD (Host Cell DNA) assay
- BCA assay
- D-Antigen assay
- SNT (Serum Neutralization Test)

Cell Culture:

- Cell line maintenance
- Cell Passaging
- Subculturing
- MTT assay
- Wound Healing assay

Molecular biology:

- SDS PAGE
- Western Blotting
- PCR
- cDNA synthesis

PROFESSIONAL EXPERIENCE:

Biological E Limited | Hyderabad

Oct 2025 – Mar 2026

Trainee | R&D-Analytical Development

- Conducted analytical characterization of 5 vaccine batch samples (PVLP) using HCP, HCD, and SDS-PAGE assays, ensuring protein quality and safety compliance with WHO and regulatory standards.
- Interpreted multi-assay experimental data to assess critical quality attributes (CQAs), supporting batch release decisions in a GMP-aligned R&D environment
- Contributed to **assay validation and regulatory compliance documentation**

Fourrts India Laboratories Pvt. Ltd| Chennai | Intern

Jan 2023 – Feb 2023

- Observed quality control, production, and formulation processes in a regulated pharmaceutical environment.
- Learned about GMP practices, documentation procedures, and workflow management in the pharmaceutical industry.

ACADEMIC CREDENTIALS:

Master of Pharmacy (Pharmaceutical biotechnology)

2024 – 2026

JSS College of Pharmacy, Ooty | CGPA-9.42/10 | Tata Trusts Scholarship

Bachelor of Pharmacy

2018 – 2022

Arulmigu Kalasalingam College of Pharmacy, Krishnankovil | CGPA-7.4/10

COURSES AND CERTIFICATES:

- E-Poster presentation on “Molecular Characterization of APOE gene SNPs in Alzheimer’s Disease using Comparative genomics and population-based analysis” and organizing committee member for international Conference “PHARMANCIA” | JSS College of Pharmacy, Ooty | 7 & 8 March 2025.
- Successfully completed the NPTEL online course and exam on intellectual property.
- Internship on Basic Cell Culture Techniques | International Research Centre, Sathyabama University | January 03-13 2025.

PUBLICATIONS:

Article titled “*Precision Probiotics for Healthy Aging: Toward Personalized Gerobiotic Therapeutics*” Under review (Manuscript submitted).

Article titled “*Molecular characterization of APOE gene SNPs in Alzheimer’s Disease*” Under review (Malaysian Journal of Medicine and Health Sciences)

LANGUAGES:

- English (Full working proficiency), Tamil (Native speaker), Telugu (Working proficiency)